GROUP 12

61.501 | PML 2026

EMGInsight

Decoding Muscle Signals with Traditional Machine Learning

PROJECT FOCUS

A shared machine learning framework for gesture recognition, finger-force estimation, and muscle-fatigue classification using public digitized sEMG datasets. We will study prediction, channel reduction, and generalization to unseen users.

COMMON ML PIPELINE

DIGITIZED EMG — Public datasets

WINDOW + FEATURES — RMS / MAV / WL

CUSTOM ML — Classification / regression

EVALUATE — F1 / MAE / RMSE

THREE COMPLEMENTARY TASKS

01 GESTURE

UCI EMG Gestures

IN 8-channel forearm EMG

OUT Hand gesture class

Assistive control / VR

02 FINGER FORCE

putEMG-Force

IN 24-channel EMG

OUT Measured force signal(s)

Prosthetic hand research

03 FATIGUE

sEMG Fatigue (Zenodo)

IN 8-channel upper-limb EMG

OUT Perceived fatigue (0/1/2)

Rehabilitation / ergonomics

RESEARCH DESIGN

Implement core traditional ML algorithms using Python/NumPy (not solely scikit-learn); compare model baselines; evaluate channel subsets within each dataset and hold out sessions/users to avoid data leakage. No deep learning.

AI FOR SOCIAL GOOD

Applications include accessible prosthetic-hand interfaces, muscle monitoring for athletes and rehabilitation research, and alternative VR interaction. Offline research only; no clinical diagnostic or device-control claims.

TEAM MEMBERS • ROLES TO BE ASSIGNED

Ma Mingyang | **Han Yuchen** | **Zhou Yin**

Wei Xin | **Gao Ruiyan** | **Li Ji**

AVAILABLE WORK PACKAGES • ALLOCATION TO BE CONFIRMED

- Gesture recognition
- Finger-force regression
- Muscle-fatigue classification
- Preprocessing & feature engineering
- From-scratch ML algorithms & testing
- Evaluation, integration & demo

DELIVERABLES

Reproducible code & custom ML implementations • Three dataset-specific studies (subject to feasibility) • Channel/model comparisons • Evaluation plots • Offline demo • GitHub and final report